

Lab 2: Description Using Models

Are Americans with higher incomes more generous?

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This study investigates the relationship between income and charitable giving proportion among US taxpayers, testing the hypothesis of a U-shaped curve where generosity is higher at lower and upper income levels. OLS regression models incorporating linear, quadratic, and logarithmic specifications for Adjusted Gross Income (AGI) were employed on county-aggregated 2022 IRS Statistics of Income data. Results differ from the hypothesis, suggesting a J-shaped positive relationship between AGI and giving proportion where a non-linear growth in charitable giving is shown with the increase in mean AGI. The preferred model, including AGI terms, dividend, capital gains, and rent/royalty, explained 78.5% of the variance in charitable contributions per return (Adjusted $R^2 = 0.785$). Findings indicate charitable generosity is complex, significantly correlated with AGI and wealth indicators in a non-linear pattern.

1. Introduction

Are Americans with higher incomes more generous than Americans with lower incomes? Giving USA's Annual Report on Philanthropy for the Year 2023 reported that individual donors gave an estimated \$374.4 billion in charitable contributions, accounting for 67% of a total of \$557.16 billion given [1]. With such a significant economic impact, a substantial body of research has sought to understand the intricacies of the relationship between an individual and their propensity to donate. Studies have explored the impact of various socioeconomic factors on charitable giving such as income inequality [2], underlying motivations such as altruism, social pressure [3] and even suggested philanthropy may be having a reverse effect on wealth redistribution [4]. As income inequality has risen dramatically over the past half century [5] a growing body of work has emerged to describe relationships pertaining to wealth distribution, such as charitable donations. Despite a unanimous consensus among researchers that the *amount* donated increases with income, research is inconsistent on how the *proportion* of income donated is related to income, as highlighted in a recent literature review of 26 studies conducted on this question [6]. As Berkeley students, we are motivated by a "commitment to contributing to a better world" [7], and by providing our findings to this question we hope

to add to an increasingly salient corpus of philanthropic research on income related giving behavior to be used by economists and policy makers seeking to mediate increasing wealth inequality.

Drawing from Neumayr & Pennerstorfer’s findings on the generosity of Americans [6], we hypothesize that the relationship between income and proportion of income donated is non-linear. Specifically, we predict a U-shaped relationship where both the lowest and highest income Americans donate a larger proportion of their income than median income Americans.

2. Description of the Data Source

The data source we have chosen to use is the IRS’ Statistics of Income (SOI) program’s ZIP Code data for the tax year 2022. This data set aggregates (at the ZIP code level) various fields within individual income tax returns filed with the IRS in 2023. There are 27,690 observations of 165 variables which the IRS samples from their master file system containing the over 200 million tax returns collected each year. Statistics are then compiled based on stratified probability samples of tax returns [8].

We recognize the following important limitations of our analysis. First, the IRS Data does not represent the full U.S. population given many individuals are not required to file an individual income tax return so we acknowledge that generalization to the U.S. population is not not feasible. Second, due to anonymity measures taken by the IRS, some US zip codes will not be represented in our study so our results equally will not be appropriate for ZIP code generalization. We omit these ZIP codes as part of our analysis which is a reduction of 102 rows. Third, although we take many steps to normalize the data, we do acknowledge that by using aggregated data we do run a risk of ecological fallacy. Finally, poorer households are more likely to take the standard deduction rather than itemize charitable donations [6] which means that charitable giving is likely to be under-reported for lower income groups.

3. Data Wrangling

To normalize the data and avoid the common pitfalls of using ZIP codes for analysis [9] we’ve chosen to merge the ZIP codes with US Census Bureau county geographies using the Housing and Urban Development’s USPS-ZIP Crosswalk file for the 4th quarter of 2022 [10]. We’ve also calculated proportional statistics (agi per return etc.) as a method of standardization. Since we are interested in individuals, not businesses, we will use the provided residential ratio (from HUD file) to allocate ZIP code level data to residential addresses taking into account both the spatial distribution of population, and the spatial distribution of residences. Although not used in our model, we added groups regarding the number of returns to explore how the relationships between our main variables of interest may differ across these groups. The final data set contains 2,983 observations with 23 variables, which is further divided into exploration (30%) and confirmation (70%) sets.

4. Operationalization

To build a robust and interpretable model, we translate theoretical concepts of income, wealth, and tax effects into measurable income based on literature related to charitable giving. Income is represented by `mean_agi_per_return`, capturing the average adjusted gross income per tax return. To account for potential non-linear effects of income, we include both a quadratic term and logarithmic transformations of `mean_agi_per_return`. The quadratic term would support modeling a potential U-shaped or inverted U-shaped relationship, while the logarithmic transformation can help normalize the data and allows us to interpret the coefficient on income as an elasticity.

Wealth and the availability of discretionary funds are operationalized using mean dividend, mean cap gain, and mean rent royalty, all divided by the number of returns, representing average income from dividends, capital gains, and rent/royalties, respectively. Although not included in our main models, stable income sources are captured by mean salary, the average salary per tax return, and to operationalize the influence of tax incentives, we include total net investment income tax amount and mean tax payment, per return.

5. Visual Description of the Relationships between Variables

The Figure 1 shows the distribution of our main variables of interest, mean AGI per return and mean charitable contribution per return. We see that both distributions have a long-right tail. The joint distribution, grouped by the number of returns shows that the relationship between these variables are approximately similar across all different groups.

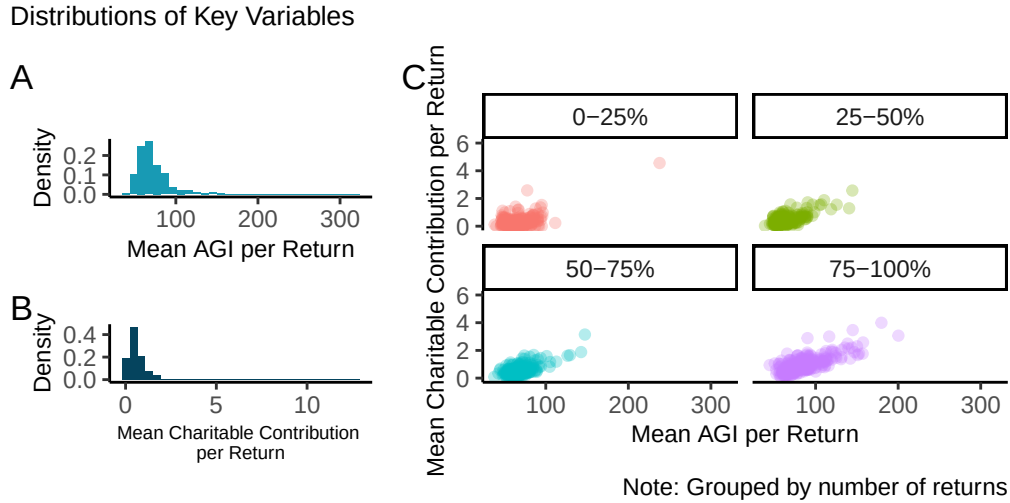


Figure 1: Panel (A) and (B) show the distribution of mean AGI and mean charitable contribution respectively. In panel (C) we present the joint distribution of the two, grouped by number of returns.

6. Model Specification

We constructed a sequence of 8 regression models including 4 main ones, incrementally adding theoretically relevant predictor variables to assess their impact on the outcome variable with different transformations. This approach allows us to sequentially examine the influence of different economic factors on philanthropic behavior per tax return. Model 1 examines a simple linear relationship. Model 2 introduces a quadratic term for mean_agi to capture potential non-linear effects. Models 4 incorporates variables that proxy for wealth and the availability of discretionary funds. Model 8 uses the log transformation with adding variables that proxy for wealth and the availability of discretionary funds. Specifications of Models 3, 5 to 7 can be found in the appendix.

Table 1: Summary Table of the Models

	Dependent Variable: mean charitable contribution			
	(1)	(2)	(4)	(8)
Mean AGI	0.027*** (0.004)	0.005*** (0.002)	0.003* (0.002)	
Mean AGI**2		0.0001*** (0.00000)	0.0001*** (0.00001)	
log(Mean AGI)				0.790*** (0.179)
Mean Dividend			0.049** (0.022)	0.072*** (0.019)
Mean Capgain			0.004 (0.003)	0.009 (0.006)
Mean Rent Royalty			0.001 (0.004)	0.001 (0.005)
Intercept	-1.316*** (0.303)	-0.240*** (0.089)	-0.335*** (0.116)	-3.393*** (0.619)
Observations	2,089	2,089	2,089	2,089
Adjusted R ²	0.602	0.706	0.785	0.721
F Statistic	3,156.028*** (df = 1; 2087)	2,512.758*** (df = 2; 2086)	1,526.683*** (df = 5; 2083)	1,350.541*** (df = 4; 2084)

Note:

*p<0.1; **p<0.05; ***p<0.01

Robust standard errors are displayed.

The covariate names omit per unit (e.g., per return) due to space limit.

7. Model Assumptions

The first assumption is that the data is IID (Independent and Identically-Distributed). The use of IRS Statistics of Income (SOI) data presents some potential challenges to this assumption. The raw data, aggregated at the ZIP code level, inherently introduces spatial dependence. While we allocate ZIP code level data to better reflect the distribution of individuals, it's still plausible that some spatial correlation stemming from socioeconomic or regional factors remains, violating independence assumption. Furthermore, while the IRS sampling process is stratified, the underlying heterogeneity of individual tax returns may still lead to concerns about the 'identically distributed' aspect. The second assumption is that a unique BLP should exist. Our visualizations show that both of our main predictor and outcome variables have a heavy right-tail. Although we see a non-infinite variance from our exploratory sample (mean AGI:4.82, mean charitable contribution:0.84), the underlying distribution may still have non-finite variance, where this assumption can be violated.

We selected Models 4 and 8 for their favorable multicollinearity scores. Model 4 uses a polynomial transformation and Model 8 employs a log transformation of mean_agi. Both models include key predictors (mean_dividend, mean_capgain, mean_rent_royalty) without problematic VIF for our main predictor (Model 4: 4.9, Model 8: 1.5). We excluded stable income and tax-related predictors, as they showed high collinearity with the mean AGI.

Model 4 showed heteroskedasticity and non-normal residuals (lighter lower, heavier upper tails). The Breusch-Pagan test for Model 4 was significant ($p < 0.001$, BP = 234.9), indicating non-constant variance. Model 8 displayed better variance consistency but still showed non-normal residuals (prominent heavier upper tail). Crucially, the Breusch-Pagan test for Model 8 was even more significant ($p < 0.001$, BP = 754), indicating a more severe issue of non-constant variance compared to Model 4, despite the visual assessment of the residual plot. Both models also exhibit non-normality in their residual distributions.

8. Model Results and Interpretation

All models significantly predict mean charitable contributions ($p < 0.001$, from F-tests). Models incorporating multiple income sources generally exhibit higher explanatory power compared to the baseline models relying solely on AGI. Model 4 includes wealth/investment income and salary, and show higher explanatory power than Models 1 and 2 (which only include AGI and AGI squared). While Model 1 explains 60.2% of the variance, Model 2 with AGI squared, increases this to 70.6%. Adding wealth/investment income (Model 4) further increases it to 78.5%. This suggests income, wealth/investment, are associated with charitable giving. Model 8 shows a 72.1% of adjusted R-squared. After assessing the linear regression assumptions and performance, we believe Model 4 as more suitable due to its balance of strong explanatory power and lesser violation of assumptions.

From model 4, we see that for every \$1,000 increase in mean dividend income, controlling for the other variables, charitable contributions are predicted to increase by \$49. Individuals with higher dividend earnings are, on average, expected to donate more, highlighting the influence of investment income on charitable giving. Higher AGI is associated with increased giving, which becomes more pronounced at higher income levels - the increase in charitable contributions associated with a \$50,000 rise in AGI is predicted to be smaller than the increase associated with the same \$50,000 rise at a higher income range (e.g., \$50,000 to \$100,000 vs \$150,000 to \$200,000).

In a broader sense, these findings indicate that charitable giving is a complex behavior influenced by a variety of predictors. Stable income, such as income from investments (dividends and capital gains) also play a role. The results do not perfectly match our initial hypothesis of a non-linear, U-shaped relationship between AGI and charitable contributions. Instead, we see a J-shaped non-linear growth which is frequently found in research. These results may be useful for charitable organizations and researchers exploring complex dynamics associated with charitable contributions, as well as policy makers aiming to enhance charitable behaviors from the public.

References

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9. Forrest, M. (2019, August 8). Stop Using Zip Codes for Geospatial Analysis. <https://carto.com/blog/zip-codes-spatial-analysis>
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Appendix

1. Link to the Data Source

- <https://www.irs.gov/statistics/soi-tax-stats-individual-income-tax-statistics-zip-code-data-soi>
- <https://www.huduser.gov/apps/public/uspsscrowswalk/login?previous=https://www.huduser.gov/apps/public/uspsscrowswalk/home>
- Link to the Google drive with the data sets downloaded from the sites above: https://drive.google.com/drive/folders/12la_DQTyAJiNlq99ZdSQrpvSMdEPky5e?usp=sharing
- Link to Github repo: https://github.com/mids-w203/Lab02_section03_c_w_f

2. List of the model specifications

Table 2: Summary Table of the Models Tried

	Dependent Variable: mean charitable contribution				
	(1)	(3)	(5)	(6)	(7)
Mean AGI	0.027*** (0.004)		-0.007 (0.005)	-0.008 (0.005)	-0.007* (0.004)
Log(Mean AGI)		-24.251*** (7.906)			
Log(Mean AGI**2)		2.985*** (0.916)			
Mean AGI**2			0.0001*** (0.00002)	0.0001*** (0.00002)	0.0001*** (0.00001)
Mean Dividend			0.054** (0.022)	0.055** (0.022)	0.056** (0.028)
Mean Capgain			0.006 (0.004)	0.006 (0.004)	0.006 (0.005)
Mean Rent Royalty			0.003 (0.004)	0.003 (0.004)	0.003 (0.004)
Mean Salary			0.012** (0.005)	0.013*** (0.005)	0.015* (0.008)
Net Investment Income Tax				-0.00000 (0.00000)	-0.00000 (0.00000)
Mean Tax Payment					-0.012 (0.040)
Intercept	-1.316*** (0.303)	49.586*** (16.995)	-0.371* (0.191)	-0.422*** (0.147)	-0.467* (0.257)
Observations	2,089	2,089	2,089	2,089	2,089
Adjusted R ²	0.602	0.644	0.790	0.792	0.792
F Statistic	3,156.028*** (df = 1; 2087)	1,891.352*** (df = 2; 2086)	1,313.285*** (df = 6; 2082)	1,135.563*** (df = 7; 2081)	994.457*** (df = 8; 2080)

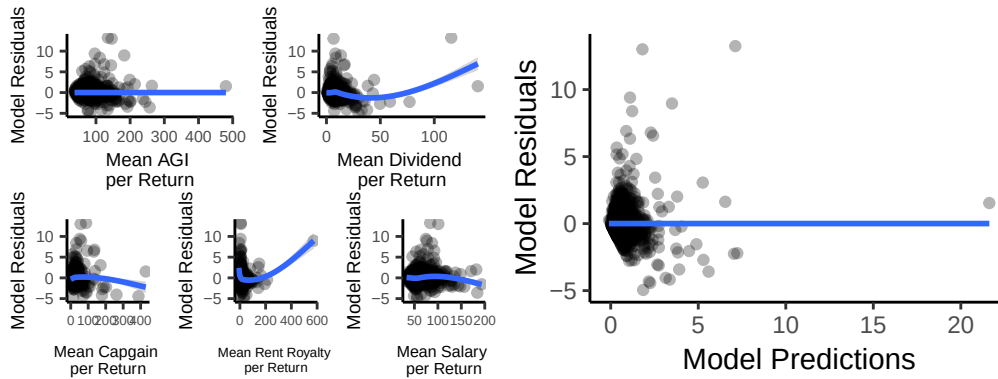
Note:

*p<0.1; **p<0.05; ***p<0.01
Robust standard errors are displayed.
The covariate names omit per unit (e.g., per return) due to space limit.

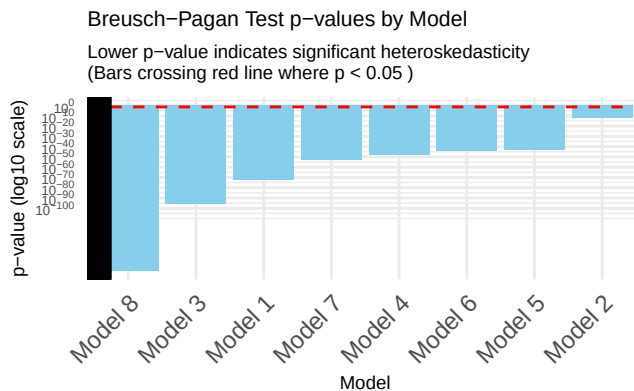
Model 3 explores a similar non-linear relationship using the natural logarithm of mean_agi and its squared term. Both coefficients are statistically significant. The adjusted R-squared of 0.644 indicates that Model 3 explains 64.4% of the variance in charitable giving, showing that even when AGI is log-transformed, it remains a significant predictor of donations. Models 5, 6, and 7 after including salary and tax-related variables reached 79.0%-79.2%. Model 6 has an adjusted R-squared of 0.792 (indicating it explains 79.2% of the variance). This shows that by adding more variables, the explanatory power of the model increases compared to the earlier models. Model 7 has an adjusted R-squared of 0.792 (indicating it also explains 79.2% of the variance). This suggests that Model 7, with its specific set of additional variables, explains a similar amount of variance as Model 6. However, these exhibit significant multicollinearity as shown by their high VIF scores, indicating their highly collinearity with the mean AGI.

3. Residuals - vs - Fitted Values Plots

Residuals vs Predictors / Fitted Values Plots



4. BP-Test Results & Model Output



Visualization of Model 4

